

A Uniform Upper Bound For Asymptotic Freeness Of $\ell_\nu^\infty(\mu)$

Automatic Math Discovery Run `fa.banach.001`

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Source And Target

The source paper is Jarno Talponen's arXiv:1604.04408, *Unconditional and bimonotone structures in high density Banach spaces*. In the notation of that paper,

$$\ell_\nu^\infty(\mu) = \ell_{<\nu^+}^\infty(\mu)$$

is the space of bounded scalar functions on μ whose supports have cardinality at most ν .

The final remarks of the source paper ask for the least cardinal $\kappa = \kappa(\lambda, \nu, \mu)$ such that $\ell_\nu^\infty(\mu)$ is (λ, κ) -asymptotically free, for infinite cardinals $\nu < \mu$ and $\lambda < 2^\nu$.

monotone basic sequence.

□

6. FINAL REMARKS

We do not know if the spaces generated by an SD sequence and with Corson's property (C) are essentially WLD in some sense. Also, we do not know, given infinite cardinals $\nu < \mu$, $\lambda < 2^\nu$, what is the least $\kappa = \kappa(\lambda, \nu, \mu)$ such that $\ell_\nu^\infty(\mu)$ is (λ, κ) -a.f. If μ is larger than ν then $\ell_\nu^\infty(\mu)$ is relatively 'spread out'. Thus this space can be viewed as being preconditioned towards a free structure alternative in a hypothetical combinatorial dichotomy. A similar cardinal invariant κ involving all reflexive Banach spaces of density $\geq \kappa$ seems reasonable. In Theorem 4.4 and Corollary 4.7 we could have managed with a Banach lattice ideal $\mathcal{I} = \mathcal{I}_{\lambda, \kappa} \subset \ell^\infty(\mu)$, in place of $\ell_\lambda^\infty(\mu)$, such that for any subspace $Y \subset \mathcal{I}$, $\text{dens}(Y) \leq \lambda$, the projection band ideal $J \subset \ell^\infty(\mu)$, generated by Y and with band projection $P_J: X \rightarrow J$, satisfies $\text{dens}(P_J(\mathcal{I})) < \kappa$. Corollary 4.7 suggests an interesting problem regarding a possible *gap* in the cardinalities of unconditional basic sequences: Is it true that each reflexive Banach space of density $\mathfrak{nc}_{\text{rf}}$ already admits an unconditional basic sequence of length ω_1 ?

The class of Banach spaces with (σ) is closed in a sense.

Proposition 6.1. *Suppose that a Banach space X is (σ) -generated, i.e. there is*

Definitions

We use the source paper's definition. Let X be a Banach space, let $\lambda < \kappa$, and let κ be an uncountable regular cardinal. The space X is (λ, κ) -asymptotically free if the following holds. Whenever $Z \subset X$ is a closed subspace with $\text{dens}(Z) \leq \lambda$, and $(Z_\alpha)_{\alpha < \kappa}$ is a nested family of closed subspaces such that

$$Z^\perp = \bigcup_{\alpha < \kappa} Z_\alpha^\perp,$$

then, for each $x \in X$,

$$\|x + z\| \geq \|z\| \quad (z \in Z)$$

implies that there is $\alpha_0 < \kappa$ such that

$$\|x + z\| \geq \|z\| \quad (z \in Z_\alpha, \alpha_0 < \alpha < \kappa).$$

Result

Theorem 1. *Let $\nu < \mu$ be infinite cardinals and let $\lambda < 2^\nu$. If κ is an uncountable regular cardinal with $2^\nu < \kappa$, then $\ell_\nu^\infty(\mu)$ is (λ, κ) -asymptotically free.*

In particular, for every regular κ with $2^\nu < \kappa \leq \mu$, this gives a positive upper-bound answer in the cardinal range explicitly asked for in the source paper's final remarks.

Idea Of The Proof

Talponen proves that $\ell_\lambda^\infty(\mu)$ is (λ, κ) -asymptotically free when $2^\lambda < \kappa \leq \mu$. The same mechanism gives a uniform bound for $\ell_\nu^\infty(\mu)$ whenever the tested subspace has density $\lambda < 2^\nu$. The point is that a dense set of size at most λ , together with one test vector, uses fewer than 2^ν coordinates. Inside that coordinate band, the density is at most 2^ν . Since this is below κ , the projected nested family of subspaces stabilizes there. The coordinate-band decomposition of the sup norm then gives the required inequality on a tail of the family.

Proof

Lemma 1. *Let A be a set with $|A| < 2^\nu$. Then*

$$\text{dens}(\ell_\nu^\infty(A)) \leq 2^\nu.$$

Proof. For each $B \subset A$ with $|B| \leq \nu$, the rational-valued functions on B , extended by zero outside B , are dense in the copy of $\ell^\infty(B)$ supported on B . Thus

$$\text{dens}(\ell_\nu^\infty(A)) \leq |[A]^{\leq \nu}| \cdot \aleph_0^\nu \leq |A|^\nu \cdot 2^\nu \leq (2^\nu)^\nu = 2^\nu.$$

□

Lemma 2. *Let κ be regular, let $P : X \rightarrow I$ be a bounded projection onto a Banach space I with $\text{dens}(I) < \kappa$, and let $(Z_\alpha)_{\alpha < \kappa}$ be a nested family of closed subspaces of X with $Z^\perp = \bigcup_{\alpha < \kappa} Z_\alpha^\perp$. Then there is $\beta < \kappa$ such that*

$$PZ_\alpha \subset \overline{PZ} \quad (\beta < \alpha < \kappa).$$

Proof. Put $E_\alpha = \overline{PZ_\alpha} \subset I$ and $E = \overline{PZ} \subset I$. If $\varphi \in E^\perp$, then $\varphi \circ P \in Z^\perp$, so $\varphi \circ P \in Z_\alpha^\perp$ for some $\alpha < \kappa$. Hence $\varphi \in E_\alpha^\perp$. Therefore

$$E^\perp = \bigcup_{\alpha < \kappa} E_\alpha^\perp.$$

If $E = I$, then there is nothing to prove. Otherwise, since $\text{dens}(I/E) < \kappa$, choose a dense family $\{u_i + E : i < \delta\}$ in the unit sphere of I/E , with $\delta < \kappa$. For each $i < \delta$, Hahn–Banach gives $\varphi_i \in E^\perp$, $\|\varphi_i\| = 1$, such that $\varphi_i(u_i) > 3/4$. Pick $\alpha_i < \kappa$ with $\varphi_i \in E_{\alpha_i}^\perp$, and let $\beta = \sup_{i < \delta} \alpha_i$. Regularity of κ gives $\beta < \kappa$.

If $y \in E_\beta \setminus E$, put $r = \text{dist}(y, E) > 0$. Choose $i < \delta$ with

$$\|y/r + E - (u_i + E)\| < 1/4.$$

Then, for some $e \in E$, $\|y/r - u_i - e\| < 1/4$. But $\varphi_i(y) = 0$ because $y \in E_\beta \subset E_{\alpha_i}$, and $\varphi_i(e) = 0$. Thus $|\varphi_i(u_i)| < 1/4$, contradicting $\varphi_i(u_i) > 3/4$. Hence $E_\beta = E$, and the conclusion follows. $\square$

Proof of Theorem 1. Let $X = \ell_\nu^\infty(\mu)$. Fix $x \in X$, a closed subspace $Z \subset X$ with $\text{dens}(Z) \leq \lambda$, and a nested family $(Z_\alpha)_{\alpha < \kappa}$ as in the definition. Assume that

$$\|x + z\| \geq \|z\| \quad (z \in Z).$$

Choose a dense set $D \subset Z$ with $|D| \leq \lambda$, and set

$$A = \text{supp}(x) \cup \bigcup_{d \in D} \text{supp}(d).$$

Each vector in X has support of cardinality at most ν , so

$$|A| \leq \nu + \lambda \cdot \nu = \max\{\nu, \lambda\} < 2^\nu.$$

Let $I = \ell_\nu^\infty(A)$, viewed as the coordinate band in X , and let $P_A : X \rightarrow I$ be the coordinate projection. The band I is closed, contains x , and contains D , hence contains Z . By Lemma 1,

$$\text{dens}(I) \leq 2^\nu < \kappa.$$

Applying Lemma 2 to P_A , there is $\beta < \kappa$ such that

$$P_A Z_\alpha \subset \overline{P_A Z} = Z \quad (\beta < \alpha < \kappa),$$

where the last equality uses $Z \subset I$ and the closedness of Z .

Now fix $\alpha > \beta$ and $z \in Z_\alpha$. Since $P_A z \in Z$, the assumption on Z gives

$$\|x + P_A z\| \geq \|P_A z\|.$$

The coordinate band decomposition in the sup norm gives

$$\|x + z\| = \max\{\|x + P_A z\|, \|(\text{Id} - P_A)z\|\} \geq \max\{\|P_A z\|, \|(\text{Id} - P_A)z\|\} = \|z\|.$$

This is exactly the required tail inequality, so X is (λ, κ) -asymptotically free. $\square$

Scope And Limitations

This is a partial result for the final-remarks question, not a computation of the exact least $\kappa(\lambda, \nu, \mu)$. It shows that the threshold $(2^\nu)^+$ is always sufficient, in the regular-cardinal regime of the definition, for every $\lambda < 2^\nu$. It gives no lower bound and leaves open whether $\ell_\nu^\infty(\mu)$ may be (λ, κ) -a.f. for some $\kappa \leq 2^\nu$.

If $\mu \leq 2^\nu$, the theorem still applies to regular $\kappa > 2^\nu$, but it gives no nontrivial instance with $\kappa \leq \mu$. This is another reason it should be read as an upper-bound packet rather than an exact answer.

Novelty And Duplicate Check

Local duplicate check: the run registry, solution index, attempts index, and proof-gap index were searched for arXiv:1604.04408, unconditional/bimonotone structures, high-density Banach spaces, density character, asymptotic freeness, and the $\ell_\nu^\infty(\mu)$ final-remarks question. No prior packet or attempt for this paper was indexed at reservation time.

Source check: the source text around Theorem 4.4 and Section 6 was inspected. Theorem 4.4 proves the special support-matched estimate for $\ell_\lambda^\infty(\mu)$ under $2^\lambda < \kappa \leq \mu$. The final remarks explicitly ask for the least $\kappa(\lambda, \nu, \mu)$ for $\ell_\nu^\infty(\mu)$ when $\lambda < 2^\nu$. The uniform 2^ν -upper bound above was not found as a stated theorem in the source.

External search on 2026-06-14 used exact phrases around arXiv:1604.04408, “asymptotically free” Banach spaces, ℓ_ν^∞ , (λ, κ) -a.f., and the source title. The only direct hit for the exact paper was the arXiv record, which lists submission on 2016-04-15 and last revision on 2016-07-07. This search is bounded rather than exhaustive, so novelty confidence is moderate.

Human Review Recommendation

The proof is short and should be checked at two points: first, the density estimate $\text{dens}(\ell_\nu^\infty(A)) \leq 2^\nu$ when $|A| < 2^\nu$; second, Lemma 2, which is the explicit form of the stabilization step used implicitly in Talponen’s proof of Theorem 4.4.

References

- [1] J. Talponen, *Unconditional and bimonotone structures in high density Banach spaces*, arXiv:1604.04408v2, 2016.